Dr Amal Chakhar

Leeds, UK | A.Chakhar@leeds.ac.uk | amalchakhar.com | ORCID 0000-0002-4294-1015 | Google Scholar

Research Interests

Remote sensing • precision agriculture • crop classification • machine learning for Earth observation • soil moisture retrieval • irrigation detection • evapotranspiration • drought monitoring.

Current Position

Postdoctoral Research Fellow, School of Earth and Environment, University of Leeds — Aug 2023 – Present

- Member of the Institute for Climate and Atmospheric Science.
- Contributing to **iSPARK** (Innovation in Sustainability, Policy, Adaptation and Resilience in Kenya): developing Earth observation methodologies for crop classification and drought and irrigation monitoring to safeguard food security in western Kenya.

Education

PhD in Agricultural and Environmental Sciences, University of Castilla-La Mancha, Spain — 2023

MSc in Integrated Planning for Rural Development and Environmental Management, Spain

Engineering Degree in Horticultural Systems, University of Sousse, Tunisia — 2015

Skills and Tools

Programming: Python (NumPy, pandas, scikit-learn), R,

Earth Observation: Sentinel-1 (SAR), Sentinel-2 (multispectral), Landsat-8, UAV photogrammetry, time-series analysis, datacubes.

Machine Learning: Supervised classification (Random Forest, SVM, kNN), accuracy assessment, feature engineering on multi-source EO data.

Geospatial: Google Earth Engine, QGIS, ArcGIS, GDAL.

Languages: Spanish, English, French (professional working proficiency); Arabic (native).

Publications

[1] **Amal Chakhar** et al. “Assessing the Accuracy of Multiple Classification Algorithms Combining Sentinel-1 and Sentinel-2 for the Citrus Crop Classification and Spatialization of the Actual Evapotranspiration Obtained from Flux Tower Eddy Covariance: Case Study of Cap Bon, Tunisia”. In: *Proceedings of IAHS* 385 (2024), pp. 443–448. DOI: 10.5194/piahs-385-443-2024.

[2] **Amal Chakhar** et al. “Irrigation Detection Using Sentinel-1 and Sentinel-2 Time Series on Fruit Tree Orchards”. In: *Remote Sensing* 16.3 (2024), p. 458. DOI: 10.3390/rs16030458.

[3] **Amal Chakhar** et al. “Optimized Software Tools to Generate Large Spatio-Temporal Data Using the Datacubes Concept: Application to Crop Classification in Cap Bon, Tunisia”. In: *Remote Sensing* 14.19 (2022), p. 5013. DOI: 10.3390/rs14195013.

[4] **Amal Chakhar** et al. “Design of a Local Nested Grid for the Optimal Combined Use of Landsat 8 and Sentinel 2 Data”. In: *Remote Sensing* 13.8 (2021), p. 1546. DOI: 10.3390/rs13081546.

[5] **Amal Chakhar** et al. “Improvement of the Soil Moisture Retrieval Procedure Based on the Integration of UAV Photogrammetry and Satellite Remote Sensing Information”. In: *Remote Sensing* 13.24 (2021), p. 4968. DOI: 10.3390/rs13244968.

[6] **Amal Chakhar** et al. “Improving the Accuracy of Multiple Algorithms for Crop Classification by Integrating Sentinel-1 Observations with Sentinel-2 Data”. In: *Remote Sensing* 13.2 (2021), p. 243. DOI: 10.3390/rs13020243.

[7] **Amal Chakhar** et al. “Applications of Remote Sensing in Precision Agriculture: A Review”. In: *Remote Sensing* 12.19 (2020), p. 3136. DOI: 10.3390/rs12193136.

[8] **Amal Chakhar** et al. “Assessing the Accuracy of Multiple Classification Algorithms for Crop Classification Using Landsat-8 and Sentinel-2 Data”. In: *Remote Sensing* 12.11 (2020), p. 1735. DOI: 10.3390/rs12111735.

[9] Rim Zitouna-Chebbi et al. “Observing Actual Evapotranspiration from Flux Tower Eddy Covariance Measurements within a Hilly Watershed: Case Study of the Kamech Site, Cap Bon Peninsula, Tunisia”. In: *Atmosphere* 9.2 (2018), p. 68. DOI: 10.3390/atmos9020068.

[10] Rim Zitouna-Chebbi et al. “Observing Actual Evapotranspiration within a Hilly Watershed: Case Study of the Kamech Site, Cap Bon Peninsula, Tunisia”. In: *Proceedings* 1.5 (2017), p. 113. DOI: 10.3390/ecas2017-04113.

Conference Presentations and Talks

Teaching and Supervision

Awards and Grants

Professional Activities

References

Available on request.